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Analytical Report 1
Pixelizing Societies from Space
April 3, 2026

Introduction

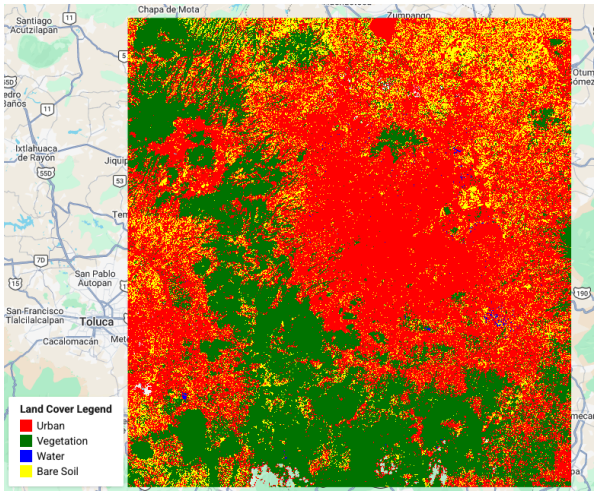
This report examines urban expansion in the Mexico City Metropolitan Area between 2005 and 2022. Mexico City is one of the largest and fastest-growing urban regions in the world, with expansion driven by population growth, economic development, and increasing demand for housing and infrastructure. The region is geographically complex, surrounded by mountains and containing a mix of urban land, vegetation, and agricultural areas. This makes it a useful case for analyzing how cities expand across different types of land cover. Using Landsat satellite imagery and supervised classification in Google Earth Engine, this study identifies where urban growth has occurred, how much land has been converted to urban use, and what spatial patterns characterize this expansion.

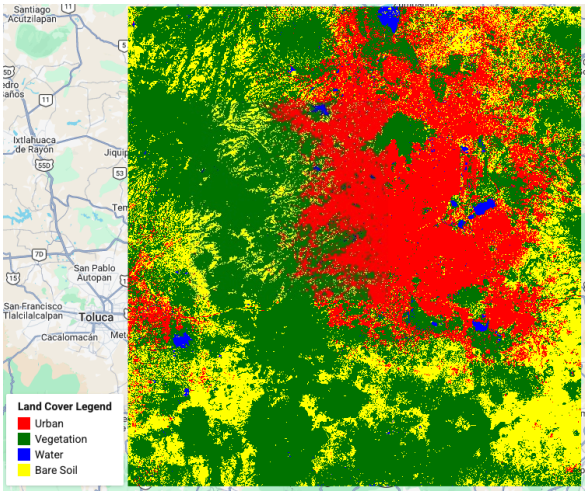
Methods

Landsat imagery was used to represent land cover conditions in 2005 and 2022. The images were filtered to reduce cloud cover and combined into composite images for each year to improve clarity and consistency. A supervised classification method was applied using a Random Forest classifier. Training data were manually selected to represent major land cover categories, including urban areas, bare soil, vegetation, and water. Separate training samples were created for each year to account for differences in how land cover appears over time. To analyze change, the classified maps from 2005 and 2022 were compared. Urban expansion was defined as areas that were classified as non-urban in 2005 and urban in 2022. Finally, the total area of this change

was calculated, and a change map was produced, where red areas represent locations of urban growth and white areas represent land that remained the same.

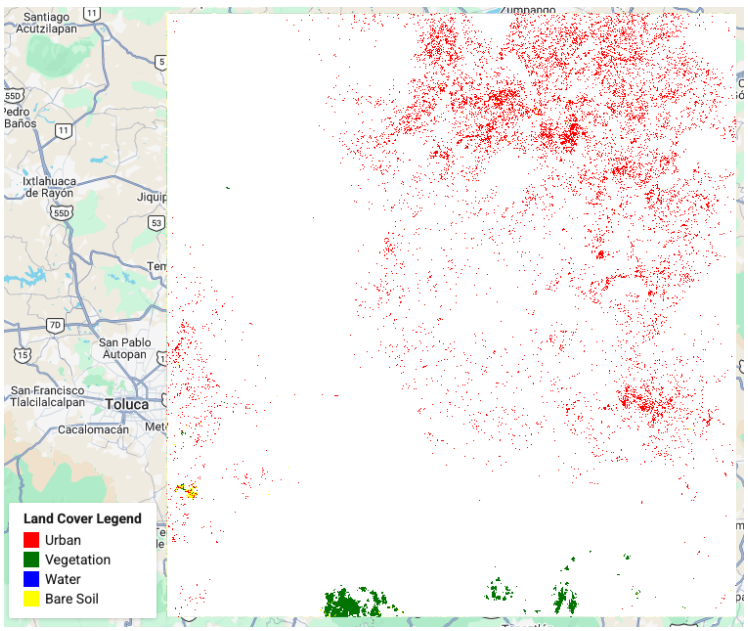
Results





The land cover maps show clear differences between 2005 and 2022. In 2005, the urban area appears more concentrated, with a distinct core surrounded by vegetation and undeveloped land (Figure 1). By 2022, the urban area had expanded significantly outward from the core (Figure 2). The increase in urban land is especially noticeable along the edges of the city, where previously non-urban land has been converted. The total urban expansion over the study period is 249.91 km², indicating substantial growth.

The pattern of urban growth is mostly continuous, meaning that the city has expanded outward from areas that were already developed. This suggests a pattern of urban sprawl. At the same time, there are some smaller and more scattered areas of growth located farther from the main city, showing that some development has happened in a more spread-out and less connected way. Most of the new urban areas appear to have replaced vegetation or other undeveloped land. This suggests that natural or semi-natural areas have been the main sources of land used for urban expansion.



Uncertainty remains due to possible classification errors, limits in training data quality, and differences in image conditions such as season and lighting. Therefore, the estimated expansion of 249.91 km² should be considered an approximation.

Conclusion

This analysis shows that the Mexico City Metropolitan Area experienced significant urban expansion between 2005 and 2022, with approximately 249.91 km² of land converted to urban use. Growth occurred mainly at the edges of the existing city, forming a largely continuous pattern of outward expansion, with some smaller, scattered developments beyond the urban core. Most of this expansion came at the expense of vegetation and other undeveloped land.

Overall, the results highlight clear patterns of urban sprawl and demonstrate how satellite imagery and classification methods can be used to track changes in land use over time. While the findings are subject to some uncertainty, they provide a useful estimate of the scale and spatial pattern of urban growth in the region.

Additional

AI tools were used to help adapt classification, organize Google Earth Engine code, and to assist in structuring the written analysis.
https://code.earthengine.google.com/70f593c3846669ae6890c53a10b73fba